

# Mission-Oriented Semantic Communications in UAV-Enabled ISAC Networks

**Abstract**—Although deep learning has considerably improved semantic communications (SemCom), image transmission systems still face persistent challenges in accurately capturing the spatial position of the person in the image and background distortion of images in unmanned aerial vehicle (UAV)-based wilderness search and rescue scenarios. To address these issues, we propose a CAFB-YOLO-based semantic communication system with a semantic knowledge base for wilderness search-and-rescue scenarios. Specifically, we design an AF module to facilitate feature fusion and multi-scale feature interaction, resulting in an improved YOLO11 model, named CAFB-YOLO. CAFB-YOLO is incorporated into the semantic communication system to improve the accuracy of capturing spatial position of the person in image. To improve the quality of the reconstructed image avoiding background distortion of images, we design a background image knowledge base, and Stable Diffusion Inpainting is adopted to generate background images for the knowledge base. To enhance transmission reliability of important semantic information, diversity combining technique, which includes maximal ratio combining (MRC), are incorporated into the system. Extensive experiments demonstrate that significantly improves image reconstruction quality, particularly under the low signal-to-noise ratio (SNR) conditions.

**Index Terms**—SemCom, YOLO, MRC, semantic knowledge base.

## I. INTRODUCTION

Unmanned Aerial Vehicle (UAV) can be remotely or automatically controlled, widely used in various fields such as military, civilian, agricultural, and environmental protection [1]. However, UAV application in outdoor human target localization remains relatively limited. Therefore, in this letter, UAV is employed for such scenarios.

In mission-oriented communication scenarios, communications resource is often limited. Due to the capability of dramatically improving information compression efficiency, semantic communications (SemCom) are introduced into UAV-enabled networks [2]. Unlike conventional communication that aims at the accurate delivery of symbols, SemCom targets the precise content reconstruction with equivalent semantics. It has the potential to achieve ultra-low compression rates and extremely high transmission efficiency.

In the field of image SemCom, existing deep learning-based research can generally be classified into two directions. The first direction relies on unimodal features, where neural network architectures are optimized to obtain high-quality semantic representations of images [3], [4]. In addition, noise-robust modules have been designed to improve the accuracy of image reconstruction under noisy channel conditions. The second direction emphasizes multimodal feature fusion, where textual and visual information are jointly exploited to achieve

richer semantic representations [5]. Compared with unimodal approaches, multimodal models generally achieve better reconstruction performance, particularly in unstable network environments. Despite achieving promising results in certain scenarios, existing semantic communication models still face two critical challenges in complex human-centric scenarios: (1) the difficulty in accurately capturing the spatial position of the person in the image, which leads to distorted or unrealistic reconstructions; (2) semantic features are vulnerable to channel noise, which results in quality degradation during image transmission. In human-centric scenarios, such as wilderness search and rescue, accurate reconstruction of the coordinate and visual appearance of the target person is essential for reliable target identification and effective rescue coordination.

Therefore, to address the first challenge, we enhance the YOLO11 detector by incorporating the proposed AF module and Triplet Attention mechanism [6], resulting in the CAFB-YOLO framework. The proposed module strengthens multi-scale feature fusion and improves the interaction between semantic and spatial information across different feature scales. As a result, CAFB-YOLO achieves superior small-object detection performance, enabling more accurate localization of outdoor human target. In addition, we incorporate the bounding box coordinate of the target person as an auxiliary modality to complement visual features. Compared with textual descriptions, the bounding box coordinate provides an explicit representation of the spatial position in human-centric images. More generally, the bounding box coordinate can distinguish the boundary between the human target and the background. These coordinates and cropped images provide explicit spatial constraints to guide semantic encoding and preserve structural consistency during transmission.

To address the second challenge, we introduce a background image knowledge base to mitigate the impact of noise on semantic features [6]. The background image knowledge base serves as a repository of the background images. By referencing a specific background image from the knowledge base, communication systems can more robustly infer and reconstruct the intended information, even when the transmitted data is corrupted or incomplete. Stable Diffusion Inpainting is adopted to generate the background images for this knowledge base. The background image knowledge base provides background information to recover the visual content of non-critical regions outside the target, therefore improving quality of reconstructed image. Considering that different semantic features have different importance, diversity combining technique MRC [7] is employed to improve reliability and spectral

efficiency in semantic features transmission.

Therefore, in this paper, we propose a CAFB-SemCom system for outdoor human target localization scenarios. Specifically, we employ CAFB-YOLO to extract semantic features and reduce redundancy. By incorporating the background image as additional auxiliary knowledge base, we achieve superior image reconstruction quality, especially under challenging channel conditions. The major contributions of this paper are summarized as follows:

- We design a CAFB-SemCom system that enables the network to more accurately capture the spatial position of the person in the image often overlooked by prior approaches, thereby achieving higher-quality image reconstruction.
- We propose a novel background image knowledge base, which differs from prior knowledge base designs based on textual annotations or category-level semantic vectors. With background information through background images, the background image knowledge base provides background priors that directly complement visual semantics and significantly enhance decoding under challenging channel conditions.

The remainder of this paper is organized as below.

## II. SYSTEM MODEL

Consider a UAV-enabled networks as shown in Fig. 2, consists of semantic encoder, transmission module, semantic decoder, and background image knowledge base. The background image knowledge base will be introduced in section III.

### A. Rician Channel

The Rician fading channel is a radio propagation model used when the received signal has both a strong direct path and reflected multipath. The equation of the Rician are described as follows

$$h = \sqrt{\frac{\kappa}{\kappa + 1}} h_{\text{LOS}} + \sqrt{\frac{1}{\kappa + 1}} h_{\text{NLOS}}, \quad (1)$$

where  $h_{\text{LOS}}$  is the LOS component and  $h_{\text{NLOS}}$  is the NLOS component that is given by the Rayleigh fading channel model,  $h_{\text{NLOS}} \sim \mathcal{CN}(0, \sigma^2)$ . Rician fading channel with additive white Gaussian noise [8] is considered as the channel model in this paper to realistically simulate the UAV-based outdoor human target localization scenarios.

### B. Semantic Encoder

The source image is encoded into three distinct semantic features [1], utilizing two semantic extractors [9] based on proposed CAFB-YOLO model, down sampling [10], and image matching algorithm. The semantic features consists of the cropped image of the target person from the source image, background image index and bounding box coordinate of the target person. The bounding box coordinate is extracted only using CAFB-YOLO model, the cropped image is extracted using CAFB-YOLO model with down sampling for further compression. For a source image  $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$ , where

$W$  and  $H$  denote the width and height of the image,  $C$  denotes the number of channels in the image, we first utilize CAFB-YOLO construct a person-masked background image  $\mathbf{U}_0 \in \mathbb{R}^{H \times W \times C}$ , where pixel values within detected human region are set to zero. Given the background image set  $\{\hat{\mathbf{U}}_1, \hat{\mathbf{U}}_2, \dots, \hat{\mathbf{U}}_N\}$ , where  $N$  denotes the number of images, we use it to query a pre-built background image index  $\alpha$ , specifically by finding the most similar image by the sum of squared differences

$$\alpha = \arg \min_i \left\| \mathbf{U}_0 - \hat{\mathbf{U}}_i \right\|_2^2, \quad i = 1, \dots, N \quad (2)$$

where  $\hat{\mathbf{U}}_i$  denotes the  $i$ -th reference background image in the background image base.

For simplicity, we use subscripts 1 and 2 to replace subscripts coordinate and cropped image of person, respectively, in the following discussion. The  $j$ -th extracted feature can be expressed by

$$\mathbf{S}_j = \mathcal{E}_j(\mathbf{X} | \theta_j^*), \forall j \in \{1, 2\} \quad (3)$$

where  $\mathcal{E}_j(\mathbf{X} | \theta_j^*)$  is the  $j$ -th model with  $\theta_j^*$  being network parameters and  $\mathbf{X}$  is the source image.

To align with existing digital communication systems, the semantic feature  $\mathbf{S}_j$  is transformed into a bit sequence  $\mathbf{b}_j$ , which can be given by

$$\mathbf{b}_j = \mathcal{B}(\mathbf{S}_j), \quad (4)$$

where  $\mathcal{B}(\cdot)$  is a binary mapping function. The background image index  $\alpha$  is mapped into a bit sequence  $\mathbf{c} = \mathcal{B}(\alpha)$ .

### C. Transmission Module

The encoded data streams are expressed as

$$\mathbf{z} = [\mathbf{b}_1, \mathbf{b}_2, \mathbf{c}], \quad (5)$$

After LDPC encoding and modulation, the transmitted symbol sequence is given by  $\mathbf{q} = [q_1, q_2, \dots, q_t]$ , where  $t$  is the number of transmitted symbols. The received data streams are given by

$$\mathbf{y} = h \odot \mathbf{q} + \mathbf{n} \quad (6)$$

where  $\mathbf{n} \sim \mathcal{CN}(0, \sigma^2)$ . After demodulation and decoding, the received semantic data streams  $\hat{\mathbf{z}} = [\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \hat{\mathbf{c}}]$ .

### D. Semantic Decoder

In the semantic decoder [11], the received semantic data streams  $\hat{\mathbf{b}}_j$  are first reconverted to the semantic features  $\hat{\mathbf{S}}_j$ , which can be expressed by

$$\hat{\mathbf{S}}_1 = \mathcal{B}^{-1}(\hat{\mathbf{b}}_1) = (\xi_1, \eta_1, \xi_2, \eta_2) \quad (7)$$

$$\hat{\mathbf{S}}_2 = \Upsilon(\mathcal{B}^{-1}(\hat{\mathbf{b}}_2)) \quad (8)$$

where  $\mathcal{B}^{-1}(\cdot)$  is the inverse operation of  $\mathcal{B}(\cdot)$ ,  $\Upsilon(\cdot)$  denotes the up-sampling operation,  $\xi_1, \eta_1, \xi_2$ , and  $\eta_2$  denote four spatial coordinate values along the horizontal and vertical dimensions of the image. The received background image

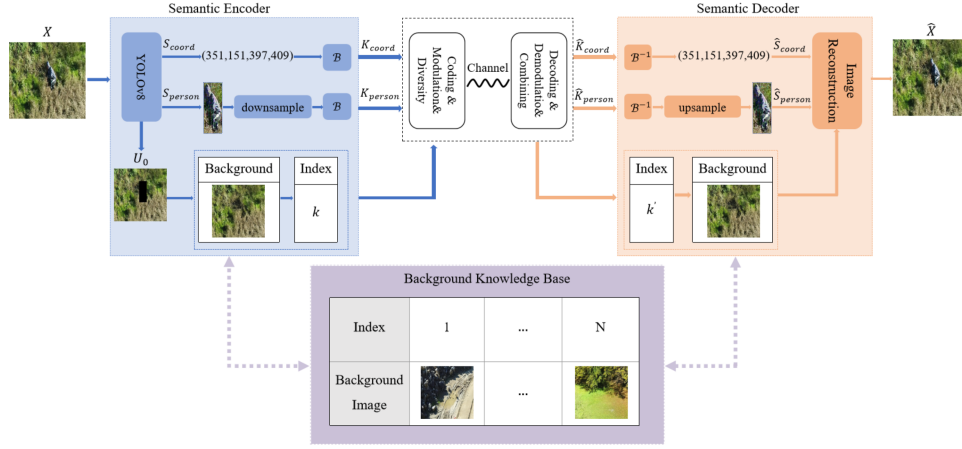


Fig. 1. The proposed semantic communication system for wilderness search and rescue scenarios.

index  $\hat{\alpha} = \mathcal{B}^{-1}(\hat{c})$ . In background image base, we use  $\hat{\alpha}$  to get a background image  $\hat{U}_0 = \hat{U}_{\hat{\alpha}}$ . Finally, the semantic features  $\hat{S}_i$  and the background image  $\hat{U}_0$  are forwarded to the image reconstruction. Let  $\Omega = \{(\xi, \eta) \mid \xi_1 \leq \xi \leq \xi_2, \eta_1 \leq \eta \leq \eta_2\}$  denote a rectangular region in the image domain. The reconstructed image  $\hat{X}$  is defined as

$$\hat{X}(\xi, \eta, \zeta) = \begin{cases} \hat{S}_2(\xi - \xi_1, \eta - \eta_1, \zeta), & (\xi, \eta) \in \Omega, \\ \hat{U}_0(\xi, \eta, \zeta), & \text{otherwise.} \end{cases} \quad (9)$$

where  $\xi, \eta$  denote spatial coordinate along the horizontal and vertical dimensions of the image and  $\zeta$  is the channel index.

### III. ALGORITHM

#### A. CAFB-YOLO

As shown in Fig. 2, the CAFB-YOLO model structure can be divided into three parts: Backbone, Neck and Head. Different from YOLO11, we improve the Neck by designing the AF module and the CAFB architecture.

In addition, the Triplet Attention mechanism introduced in Reference [7] and the SDI module are incorporated into the proposed model structure.

Reference [7] introduces the Triplet Attention mechanism in the neck feature fusion layer. Different from the module that only focuses on channel information or the Spatial Attention that only focuses on spatial information, the Triplet Attention establishes cross-dimension interaction dependencies through three branches without dimensionality reduction operations, thus avoiding the loss of feature information.

However, the performance improvement achieved by directly applying Triplet Attention to the YOLO11 model is not significant. Therefore, based on this mechanism, we further design the CAFB architecture. Specifically, four feature layers from the Backbone are introduced into the Neck, and the proposed AF module is employed to achieve feature fusion and the interaction of multi-scale feature information.

The proposed AF module is shown in Fig. 3, AF integrates channel information and spatial information from three feature scales, three input feature are represented as

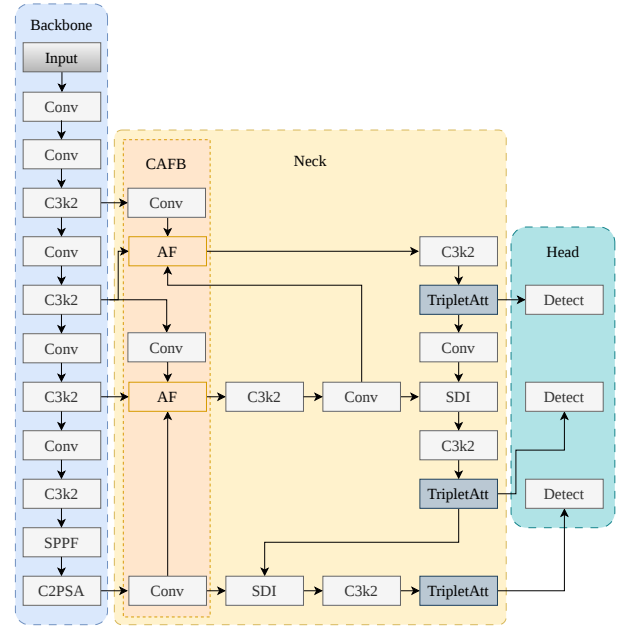


Fig. 2. The CAFB-YOLO model structure.

$\mathbf{X}_l \in \mathbb{R}^{C_l \times H_l \times W_l}$ ,  $\mathbf{X}_m \in \mathbb{R}^{C_m \times H_m \times W_m}$ ,  $\mathbf{X}_s \in \mathbb{R}^{C_s \times H_s \times W_s}$ , where  $\mathbf{X}_l$  denotes the large-scale feature,  $\mathbf{X}_m$  denotes the middle-scale feature,  $\mathbf{X}_s$  denotes the small-scale feature. After processing through the AF module, the feature representation can be expressed as

$$\mathbf{F} = \text{Concat}(\mathcal{P}_{\max}(\mathbf{X}_l) + \mathcal{P}_{\text{avg}}(\mathbf{X}_l), \mathbf{X}_m, \mathcal{U}(\mathbf{X}_s)) \quad (10)$$

where  $\mathcal{P}_{\max}(\cdot)$  denotes the adaptive maximum pooling operation,  $\mathcal{P}_{\text{avg}}(\cdot)$  denotes the adaptive average pooling operation,  $\text{Concat}(\cdot)$  denotes fusion in the channel direction, and  $\mathbf{F} \in \mathbb{R}^{C_m \times H_m \times W_m}$ .

By integrating high-level semantic information, middle-level information, and low-level spatial details, the proposed module effectively enhances multi-scale feature representation capabil-

ity, thereby improving detection performance, especially for small objects and wilderness scenes.

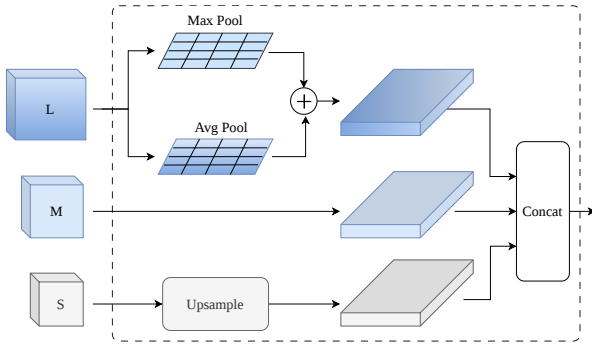


Fig. 3. The AF module.

### B. Background Image Knowledge Base

The proposed background image knowledge base is applied to the reconstruction of target person images in wilderness search and rescue scenarios. The background image knowledge base provides a background image as input to the image reconstruction of semantic decoder.

Stable Diffusion Inpainting is a latent text-to-image diffusion model [12] that can generate photo-realistic images from text input and can also perform inpainting by using a mask.

In this paper, the Stable Diffusion Inpainting is used to generate a background image  $U$  in the background image knowledge base. To improve its performance, several improvements are applied to the model inputs. The Stable Diffusion Inpainting model requires three inputs, including a text prompt, an input image, and a corresponding mask image. The text prompt can be manually set to an appropriate value according to the wilderness search and rescue scenarios, and it has little impact on the output  $U$ . The mask image is obtained by directly processing the input image with a pretrained background removal model, namely rembg.

## IV. EXPERIMENTS

We conducted experiments on the SARD dataset, with a resolution of  $640 \times 640$ . The experiments include the training and testing of AF-YOLO model, as well as simulations of the semantic communication system.

### A. AF-YOLO

The original SARD dataset contains 5755 images of wilderness search and rescue scenarios. 4041 images were selected as the training set, 1144 images as the validation set and 570 images as the test set.

We train detectors with GPU (NVIDIA RTX 3050), adopting the stochastic gradient descent (SGD) optimizer with the initial learning rate 0.001, and cosine annealing method is used to reduce the learning rate. The batch size is 16, the total number of training rounds (epochs) is 100

In order to evaluate the detection effectiveness of AF-YOLO model algorithm, the following evaluation metrics are

TABLE I  
PERFORMANCE VERSUS THE MODEL.

| Model          | mAP@0.5      | mAP          | Inference Time (ms) |
|----------------|--------------|--------------|---------------------|
| YOLO11         | 0.918        | 0.553        | 8.9                 |
| TripleAtt-YOLO | 0.908        | 0.554        | 11.0                |
| AF-YOLO        | <b>0.923</b> | <b>0.581</b> | <b>12.5</b>         |

used in this paper: mean Average Precision(mAP@50-95 and mAP@50), and Inference Time. A comparison was conducted between AF-YOLO and the smaller series of models within the YOLO11 series. The results are summarized in Table I, it is evident that AF-YOLO surpasses YOLO11 and TripletAtt-YOLO in terms of accuracy while maintaining comparable inference speed with TripletAtt-YOLO.

### B. Semantic Communication System Performance

Semantic communication system simulations are conducted on the test set consisting of 570 images. We evaluated the model on Rician channel. The SNR was varied from 0 dB to 16 dB to simulate different channel conditions. The modulation and coding schemes of these two semantic data streams are different. The semantic data stream  $K_1$  employs the diversity and combining techniques MRC with 3 receiving branches with  $L = 3$ , whereas  $K_2$  does not. The coordinate data streams employ LDPC coding with a rate of 1/4 and BPSK modulation, but the cropped person image data streams employ LDPC coding with a rate of 1/4 and BPSK modulation. Monte Carlo simulation approach is used in simulation of transmission stage.

To evaluate the performance of the proposed model, we compare it with several baseline methods, including DynamicJSCC [1], CAFB-SemCom, DeepJSCC [7], and the traditional JPEG+LDPC pipeline. We adopt three widely used metrics to quantitatively assess the quality of the reconstruction images: LPIPS, SSIM, and PSNR. Specifically, PSNR quantifies pixel-level reconstruction fidelity, SSIM measures structural similarity, and LPIPS evaluates perceptual similarity based on deep neural network features. In addition, we report the runtime latency to assess the practical feasibility of the proposed framework.

Figs. 4 compares the performance of the proposed method with baseline schemes over an Rician channel in terms of PSNR, SSIM, and LPIPS. As shown, CAFB-SemCom consistently outperforms all baseline models across the entire SNR range. This performance gain is mainly attributed to the incorporation of person position information and background information, which enables more accurate modeling of image spatial structure. Moreover, the improvement is more pronounced under low-SNR conditions, where the background image knowledge base of the CAFB-SemCom provides prior background images information that enhances robustness against channel noise.

Figs. ??-?? presents a qualitative comparison of the image reconstruction results under various SNRs for the coded mod-

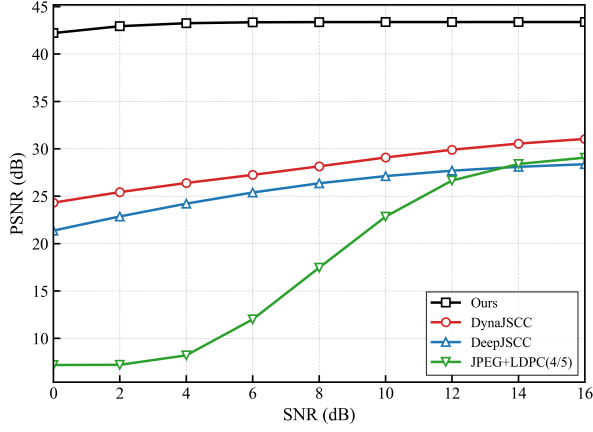


Fig. 4. The BER performance of the cropped person image under different coded modulation schemes.

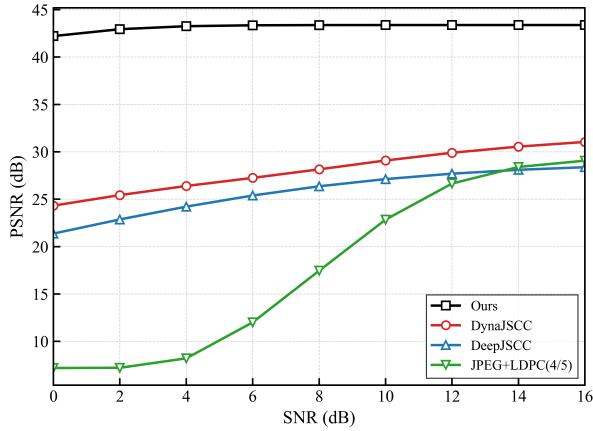


Fig. 5. The BER performance of the total under different coded modulation schemes.

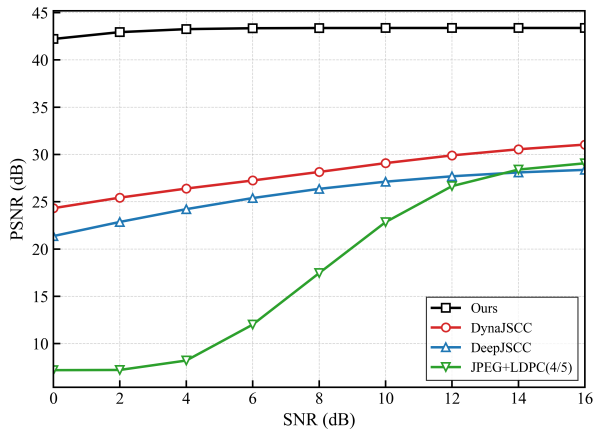


Fig. 6. The BER performance of the total under different combining strategies.

ulation scheme CMSn. It can be observed that all six coded modulation schemes achieve relatively high image quality. Even at low SNR, the target can still be accurately localized, which is of great importance in wilderness search and rescue scenarios.

## V. CONCLUSION

This letter proposes a semantic communication system model named CAFB-SemCom for image transmission. By incorporating CAFB-YOLO model, CAFB-SemCom is able to better capture structural information of the image, thereby extracting more meaningful semantic features. The CAFB-SemCom system transmits only the critical semantic information, while the remaining non-essential information is recovered at the receiver based on prior knowledge stored in a knowledge base, simplifying the communication process and reducing storage requirements. By introducing diversity and combining technique MRC, our system effectively improves transmission of critical semantic information at higher data rate and certain transmission quality, while decreasing the BER of person cropped images, thereby improving reconstruction image quality. Simulation results demonstrate that the CAFB-SemCom system employing the proposed CA achieves quality improvements of image reconstruction UAV-based wilderness search and rescue scenarios. For wilderness search and rescue scenarios, the target can still be accurately localized even in challenging channel with significant noise interference.

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